

Class 14

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```
library(readr)
```

Warning: package 'readr' was built under R version 4.5.2

```
b <- read_tsv("mm-second.x.zebrafish.tsv")
```

Rows: 39829 Columns: 12

-- Column specification -----

Delimiter: "\t"

chr (2): NP_034603.2, XP_002663941.1

dbl (10): 44.444, 90, 46, 1, 644, 733, 295, 380, 6.07e-20, 94.4

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

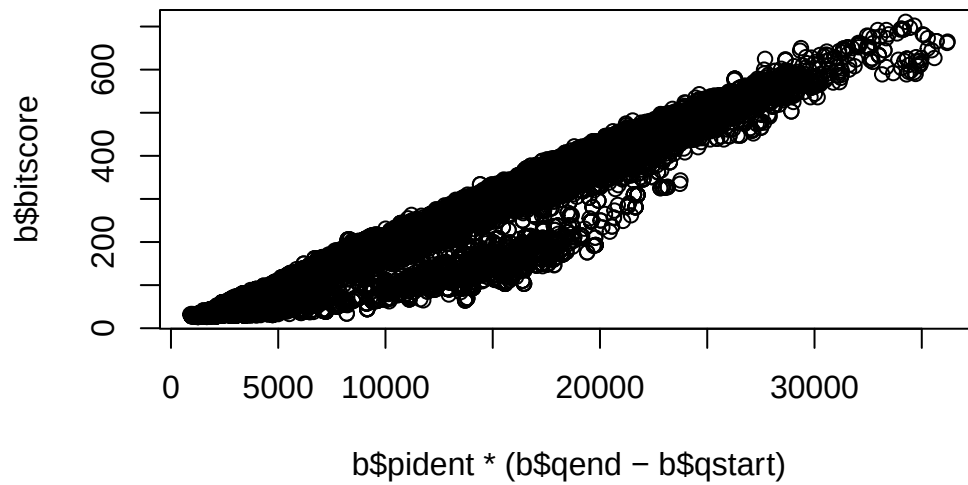
```
colnames(b) <- c("qseqid", "sseqid", "pident", "length", "mismatch", "gapopen", "qstart", "qend", "sstart", "send")
head(b)
```

A tibble: 6 x 12

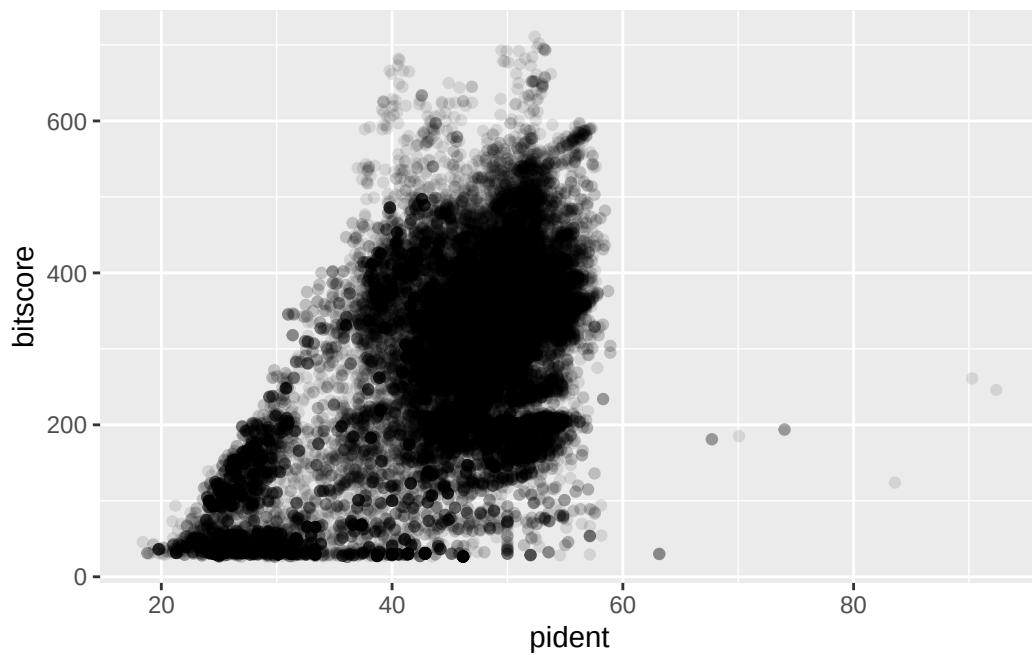
	qseqid	sseqid	pident	length	mismatch	gapopen	qstart	qend	sstart	send
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	NP_034603.2	XP_02133~	41.7	96	46	2	644	733	295	386
2	NP_034603.2	XP_02132~	43.2	74	41	1	660	733	217	289
3	NP_036084.2	XP_07379~	28.2	209	140	6	17	220	1	204
4	NP_036084.2	NP_00115~	25.9	205	143	3	27	224	24	226
5	NP_036084.2	XP_07378~	24.3	136	97	1	87	216	18	153
6	NP_036084.2	XP_06807~	26.6	143	98	3	87	224	112	252

i 2 more variables: evalule <dbl>, bitscore <dbl>

```
## Assuming your blast results are stored in an object called 'b'  
plot(b$pident * (b$qend - b$qstart), b$bitscore)
```



```
library(ggplot2)  
ggplot(b, aes(pident, bitscore)) + geom_point(alpha=0.1)
```



```
ggplot(b, aes((b$pident * (b$qend - b$qstart)), bitscore)) + geom_point(alpha=0.1) + geom_smooth()
```

`geom\_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'

